Study of Muon Detector Response of GRAPES-3 experiment using Cosmic Ray Induced Extensive Air Showers

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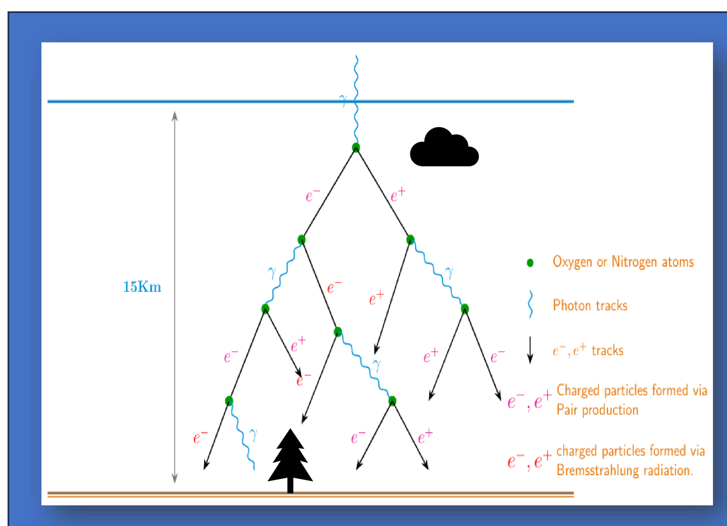
Motivation

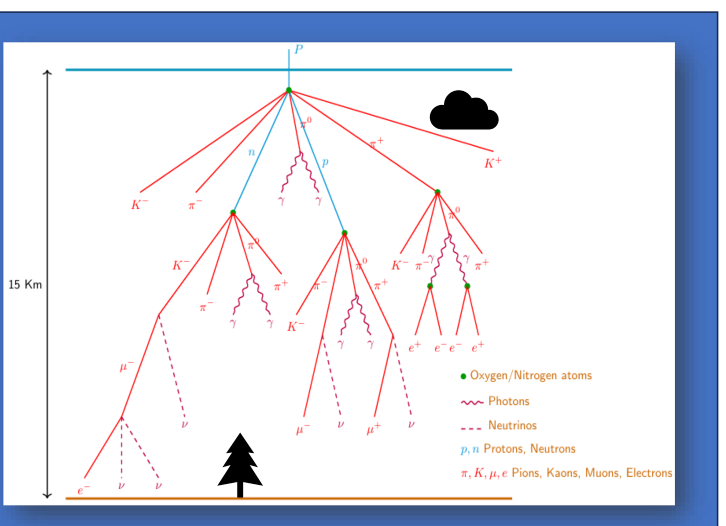
- Cosmic rays probe extreme astrophysical environments at energies beyond current terrestrial accelerators
- Understanding their origin and acceleration mechanisms remains a fundamental a open problem

Objective

- Study the physics of Extensive Air Showers initiated by cosmic rays
- Perform timing analysis on the world's largest muon telescope, GRAPES-3
- Muon multiplicity is sensitive to the composition of the primary cosmic ray particle
- Determine the muon multiplicity using machine learning approaches to infer primary particle composition



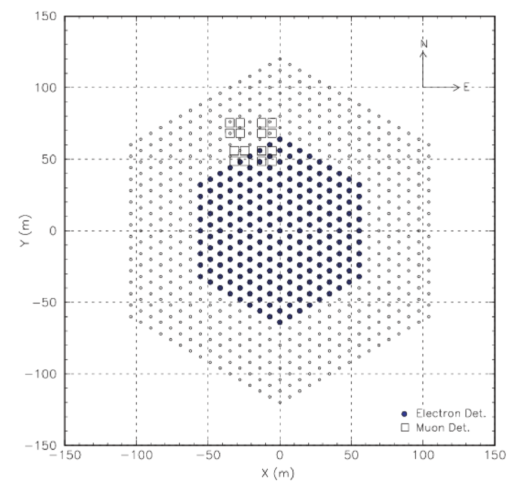
Gamma Initiated Air Showers

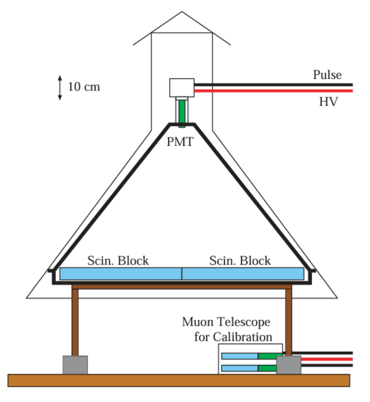


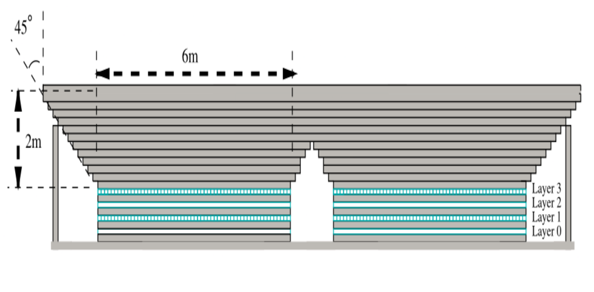
Proton Initiated Air Showers

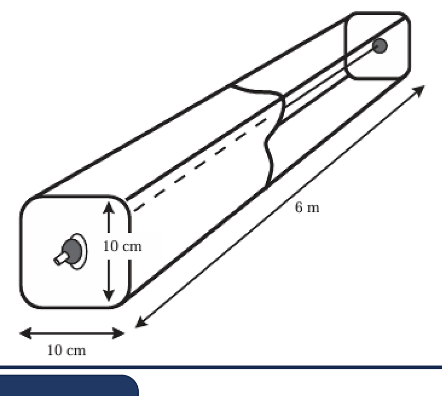
Gamma Ray Astronomy at PeV EnergieS Phase -3

- Comprises two main components
  - Shower Array Detector consisting scintillator detector
  - Muon Telescope consisting proportional counter
- Trigger system
  - EAS trigger - 2 Level trigger for shower array
    - L0 – requires any 3 consecutive layers in the array to get triggered withing 100 ns window
    - L1 – requires n out of 127 detectors to get triggered within 1 µs window
  - 4-Fold trigger – all 4 layers to be triggered withing 3 µs

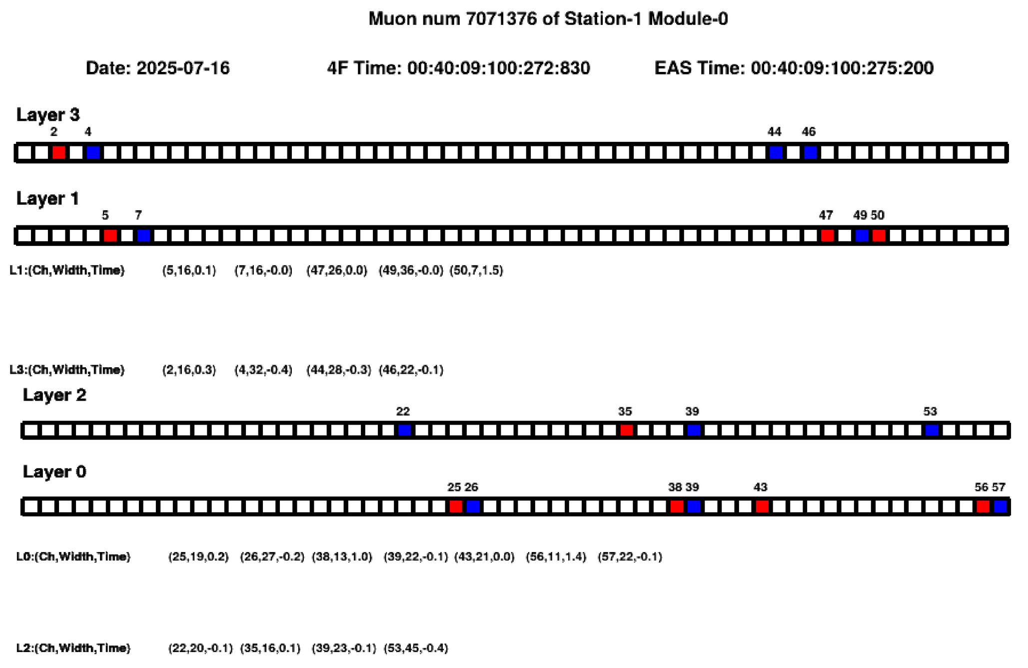






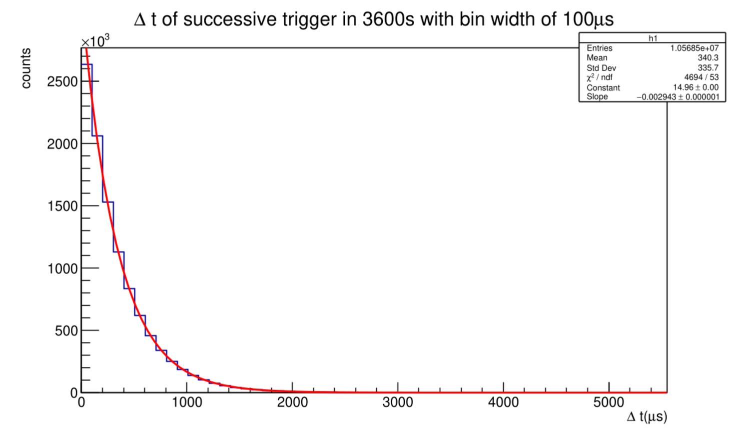


Methodologies

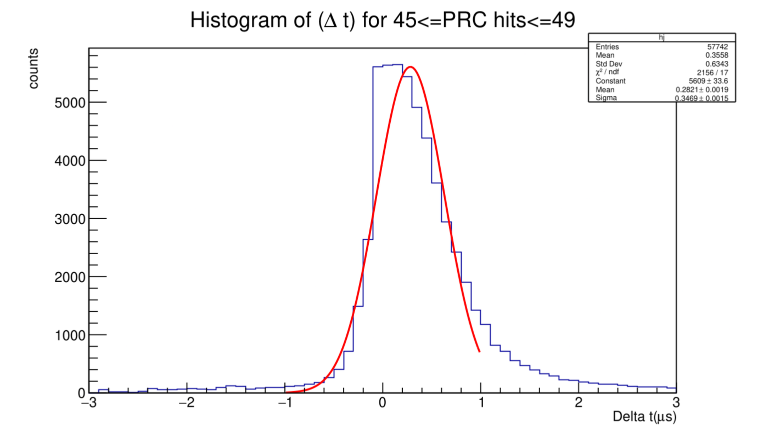


- Event display software developed in this work was used to record events
- Determine muon multiplicity corresponding to EAS and 4Fold trigger
- Match muon track angle with air shower
- Differentiate the type of shower based on muonic-content

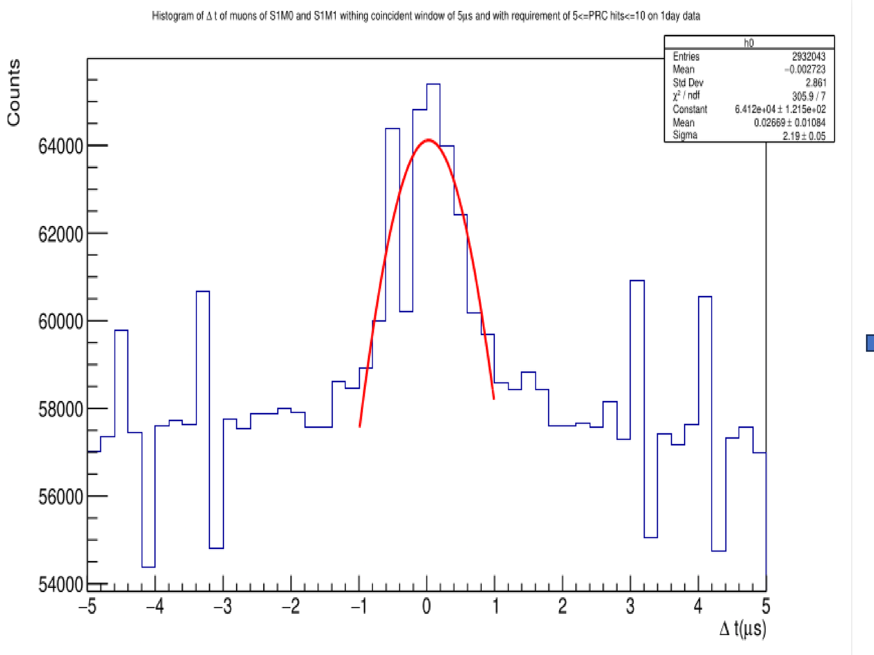
Timing analysis



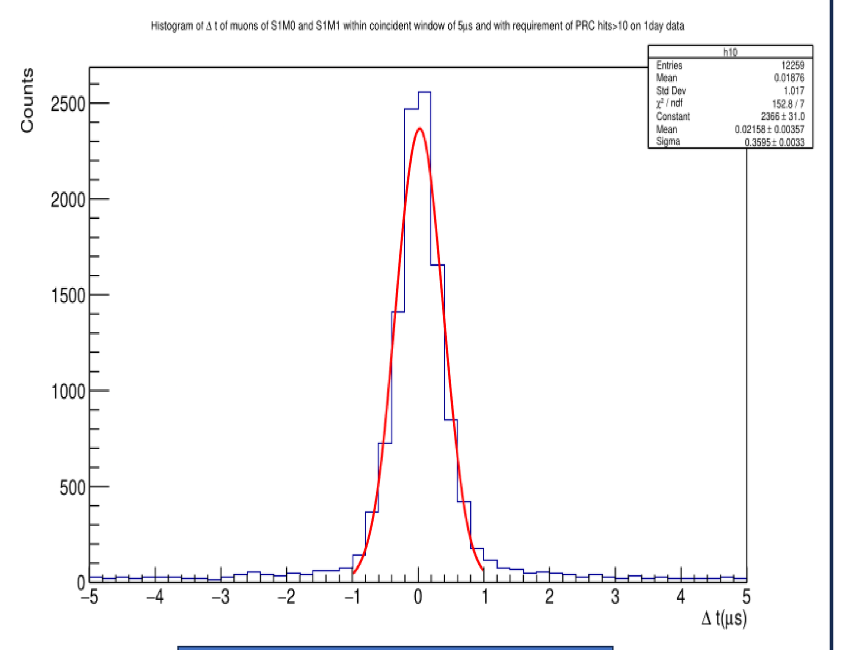
Nature of Muon arrival



Response time

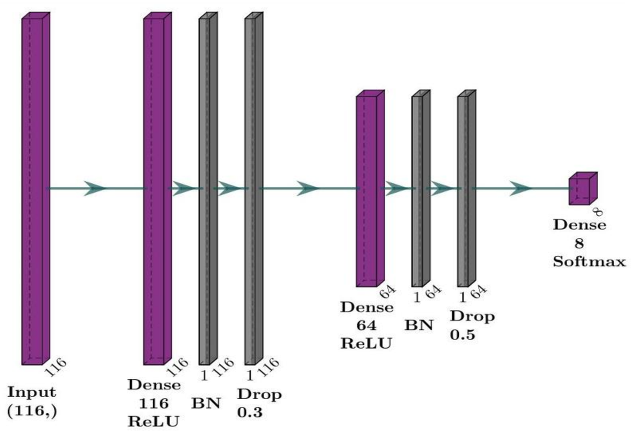


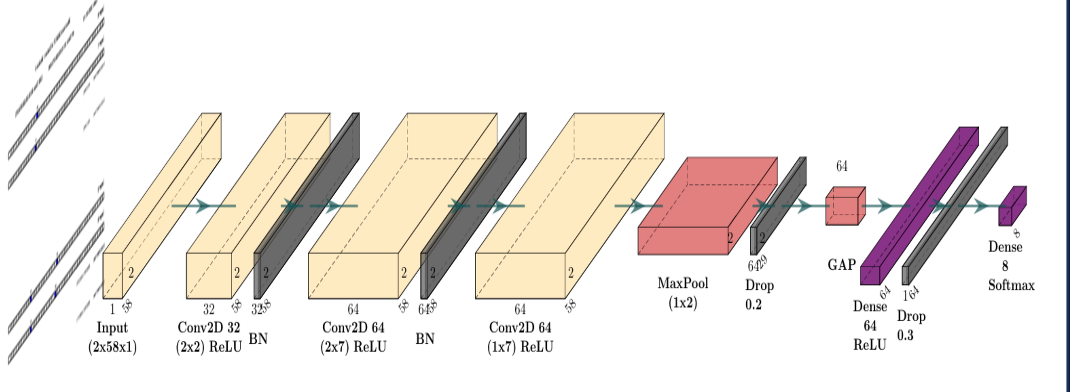
Response of 2 modules



Eliminating uncorrelated muons

Machine Learning approaches





$$z = \sum_{k=1}^{m} w_k x_k + b \;\Rightarrow\; y = f(z) \;\Rightarrow\; E = \frac{1}{2n}\sum_{i=1}^{n}(t^i - y^i)^2$$

$$G_k = \frac{\partial E}{\partial w_k}$$

$$\frac{\partial E}{\partial w_{ij}} = y_i y_j (1 - y_j)(t_j - y_j)$$

$$\Delta w_k = -\varepsilon G_k$$

Results and Inferences

		Macro Recall for Datasets					
		DS1_orig	DS2_orig	DS3_orig	DS1_syn	DS2_syn	DS3_syn
FCNN	Normal	0.41	0.24	0.17	0.68	0.67	0.35
	Class weights	0.38	0.26	0.19	0.75	0.72	0.46
	Focal loss	0.34	0.27	0.18	0.71	0.62	0.43
	Balanced batches	0.41	0.28	0.18	0.82	0.74	0.45
CNN	Normal	0.64	0.68	0.44	0.84	0.83	0.49
	Class weights	0.62	0.68	0.59	0.83	0.83	0.76
	Focal loss	0.56	0.7	0.68	0.84	0.85	0.85
	Balanced batches	0.64	0.53	0.61	0.84	0.89	0.73

- Muon arrival follows Poisson statistics with a mean time interval of 340 µs
- Mean response time of the proportional counter is 400 ns
- CNN model performs best across all three datasets, achieving **macro-recall ~ 85%**
- Individual class recall exceeds **90%**, except for extreme minority classes

References and Acknowledgement

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